

Run 06 | NME

India

IV only

Target Necessity-motivated entrepreneurship; binary 1 vs. 0

Sample 27,555 respondents **Coverage** 1 country; 2015–2022

Positive cases 1,849 (6.71%) **Case time** 37s

Train / val. / test 18,740 / 3,305 / 5,510 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9801). Highest F1: H2O GBM (0.7222). The all-negative accuracy baseline is 93.29% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9801	0.6823	95.55	62.40	84.55	0.7181	<1s
XGBoost	0.9796	0.6795	95.39	60.67	88.62	0.7203	<1s
H2O GBM	0.9795	0.6771	95.15	58.61	94.04	0.7222	2s
CatBoost	0.9787	0.6642	95.26	60.19	86.45	0.7097	3s
Ridge	0.9773	0.6523	94.19	53.87	92.41	0.6806	8s
Decision Tree	0.9770	0.6218	94.41	54.70	96.21	0.6974	<1s
Logistic Regression	0.9770	0.6387	94.17	53.73	93.77	0.6831	<1s
Elastic Net	0.9769	0.6342	94.32	54.36	94.58	0.6904	9s
Lasso	0.9767	0.6340	94.48	55.15	94.31	0.6960	8s
Naive Bayes	0.9096	0.4146	92.40	45.79	73.71	0.5649	3s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	705.81	52.09
2	Established business owner	290.26	21.42
3	Good career choice	39.38	2.91
4	Knowing entrepreneurs	31.12	2.30
5	Self-efficacy	29.37	2.17
6	Media coverage	20.47	1.51
7	Business angel	20.07	1.48
8	Business exit (missing indicator)	19.60	1.45

Share is the percentage of total forest impurity importance across all 25 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 06_obj1_india_iv_NME. Scores and importance are taken directly from saved CSV outputs.